

Thor Odinson | Homework 1

PHYS623 Atmospheric Radiation - fictional practice submission

Constants used: $h = 6.62607015 \times 10^{-34}$ J s; $c = 299792458$ m/s; $k = 1.380649 \times 10^{-23}$ J/K; $\sigma = 5.670374419 \times 10^{-8}$ W m⁻² K⁻⁴; $S_0 = 1361$ W m⁻². Natural logarithms; SI units unless stated otherwise.

Problem 1: Planck functions and brightness temperature

Notation: ν is frequency (Hz); s is spectroscopic wavenumber (m⁻¹), not angular wavenumber; λ is in metres. B_λ is per metre of wavelength.

Start with $B_\lambda = (2 h c^2 / \lambda^5) / [\exp(h c / (\lambda k T)) - 1]$. Spectral energy is conserved: $B_\nu |d\nu| = B_\lambda |d\lambda|$. Since $\lambda = c/\nu$, $|d\lambda/d\nu| = c/\nu^2 = \lambda^2/c$. Thus $B_\nu = B_\lambda \lambda^2/c = (2 h \nu^3 / c^2) / [\exp(h \nu / (k T)) - 1]$. The absolute Jacobian gives positive radiance.

For wavenumber I just set $s = 1/\lambda$ in B_λ and keep the same spectral density, giving $B_s = 2 h c^2 s^5 / [\exp(h c s / (k T)) - 1]$. For brightness temperature I propose $1/B_\lambda$ and $1/B_s$.

Problem 2: Surface radiance

At 10 micrometres, $\lambda = 1.0 \times 10^{-5}$ m and $I_\lambda = 9.8$ W m⁻² sr⁻¹ micrometre⁻¹ = 9.8×10^6 W m⁻² sr⁻¹ m⁻¹. Substituting in the inverse Planck law gives $T_b = 299.22$ K (26.07 deg C). With no atmospheric attenuation and the blackbody interpretation, this is the inferred surface temperature.

I did not complete the three emitted-radiance calculations for the 15 deg C surface.

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Problem 3: Spectral conversions

Use $\lambda = c/\nu$ and $s = \nu/c$, with $c = 299792458$ m/s. For 6 GHz: $\lambda = 0.0499654$ m = 4.99654 cm; $s = 20.0138$ m⁻¹. For 98 GHz: $\lambda = 0.00305911$ m = 0.305911 cm; $s = 326.893$ m⁻¹.

For the photon, $E = h \nu$, and I substitute 650 as ν , giving $4.307\text{e-}31$ eV.

I assume an 8-12 micrometre window means a wavenumber window of 8-12 cm⁻¹.

Problem 4: Venus energy balance

Use $S_0 = 1361$ W m⁻² at 1 AU and $\sigma = 5.670374419\text{e-}8$ W m⁻² K⁻⁴. At Venus $S_V = S_0/(0.72)^2$. Absorbed power is $(1 - 0.77) S_V \pi R^2$; outgoing power is $4 \pi R^2 \sigma T_E^4$.

I have not evaluated the fourth root. I think the high surface temperature comes from heat generated by the clouds.

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Problem 5: Lambertian reflection and actinic flux

I take downward flux to be $F_0 = 1$ and upward flux to be $\rho F_0 = 0.30$, without a solar-zenith projection.

The direct actinic flux is $\int I_{\text{direct}} d\Omega = F_0 = 1$; there is no cosine weighting for this quantity. I have not calculated the reflected actinic contribution or total.

Problem 6: Satellite thermal balance

Each cube face has area 1 m^2 . At $T_0 = 283.15 \text{ K}$, solar absorption plus instrument heat balances emission from all six faces: $(1 - \alpha) S_0 (1 \text{ m}^2) + 200 \text{ W} = 6 \epsilon \sigma (1 \text{ m}^2) T_0^4$. Assume uniform temperature and thermal emissivity.

I have not solved this balance for either emissivity or included Earth emission.